



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- data wrangling
- exploratory data analysis (EDA) using visualization and SQL
- interactive visual analytics using Folium and Plotly Dash
- predictive analysis using classification models

Summary of all results

- data wrangling results
- EDA using visualization and SQL results
- interactive visual analytics results
- predictive analysis results

Introduction

- Project background and context

SpaceX's accomplishments include: Sending spacecraft to the International Space Station. One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. SpaceX's Falcon 9 launch like regular rockets. The payload is enclosed in the fairings. Stage two, or the second stage, helps bring the payload to orbit, but most of the work is done by the first stage. Unlike other rocket providers, SpaceX's Falcon 9 Can recover the first stage. Sometimes the first stage does not land. Other times, Space X will sacrifice the first stage due to the mission parameters like payload, orbit, and customer.

- Problems you want to find answers

Your job is to determine the price of each launch. You will do this by gathering information about Space X and creating dashboards for your team. You will also determine if SpaceX will reuse the first stage. Instead of using rocket science to determine if the first stage will land successfully, you will train a machine learning model and use public information to predict if SpaceX will reuse the first stage.

Section 1

Methodology

Methodology

Executive Summary

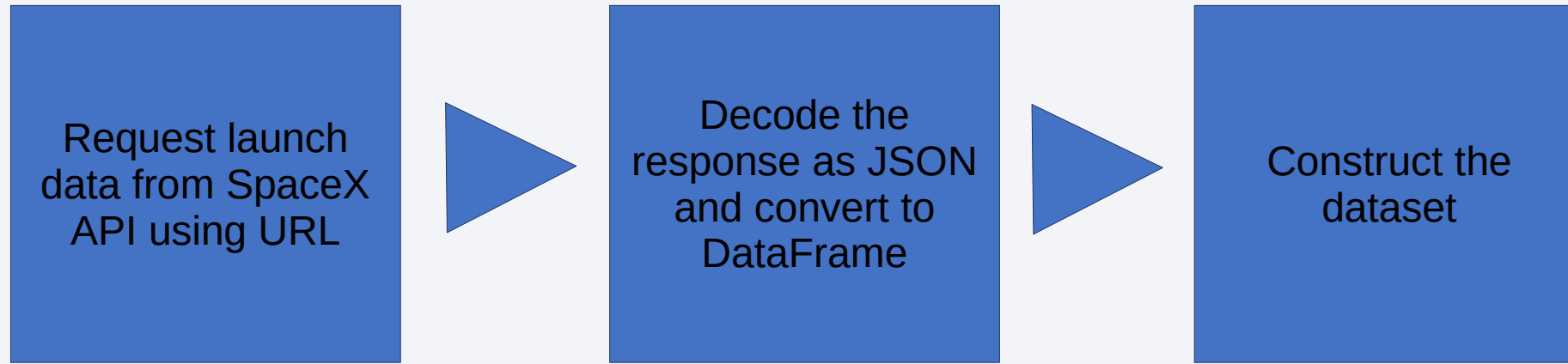
- Data collection methodology:
 - Collection via API request and webscraping (BeautifulSoup)
- Perform data wrangling
 - Manipulation via Pandas
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Initializing models, fitting and tuning, predicting

Data Collection

Data was collected by using `get_request` to SpaceX API and web scraping using BeautifulSoup.

Decoded html as json and fed into pandas dataframe.

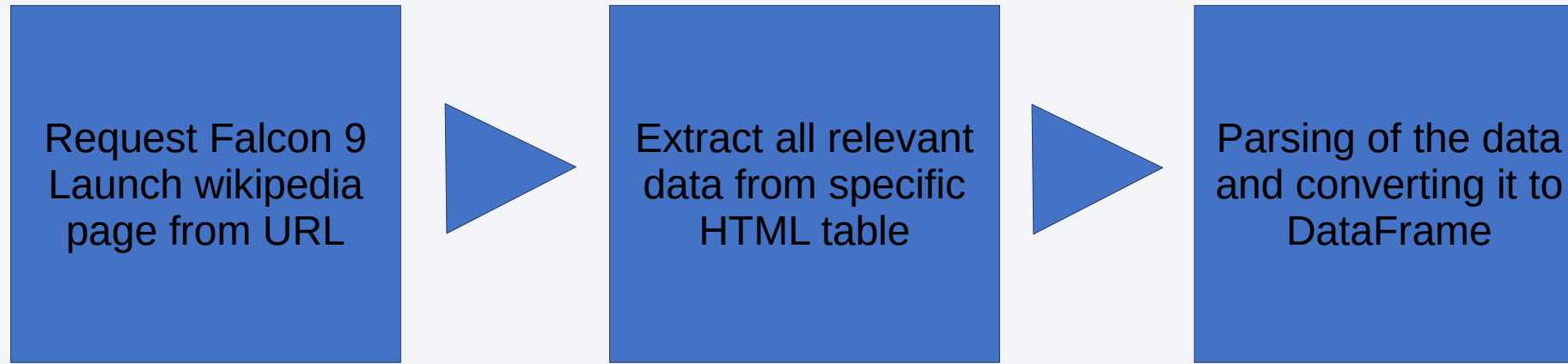
Data Collection – SpaceX API



- Github:

[https://github.com/m150x/
ibm_data-science/blob/main/
capstone/jupyter-labs-spacex-
data-collection-api.ipynb](https://github.com/m150x/ibm_data-science/blob/main/capstone/jupyter-labs-spacex-data-collection-api.ipynb)

Data Collection - Scraping



- Github:

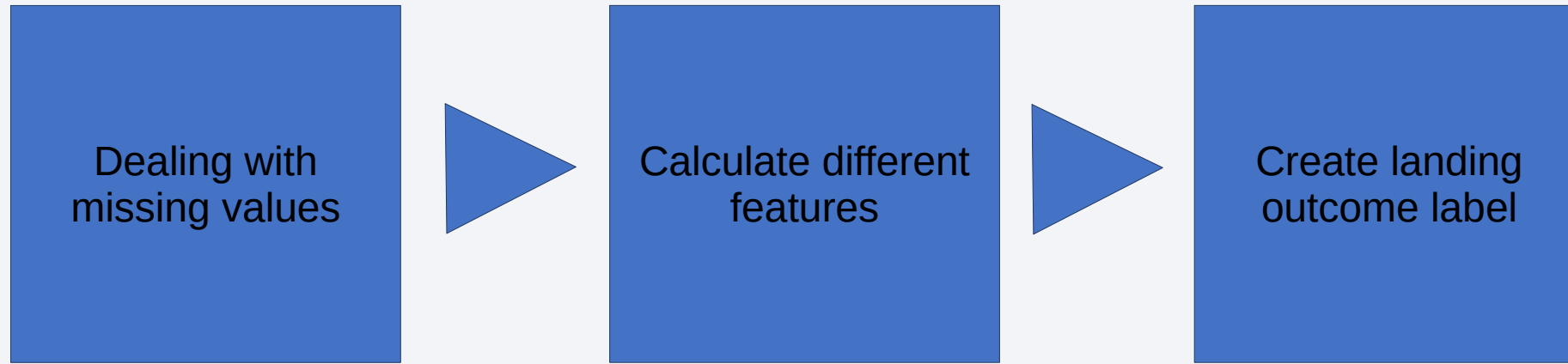
https://github.com/m150x/ibm_data-science/blob/main/capstone/jupyter-labs-webscraping.ipynb

Data Wrangling

First impressions of newly DataFrame (isnull, dtypes, etc.)

Create a landing outcome label.

Data Wrangling



- Github:

[https://github.com/m150x/
ibm_data-science/blob/
main/capstone/labs-jupyter-
spacex-Data
%20wrangling.ipynb](https://github.com/m150x/ibm_data-science/blob/main/capstone/labs-jupyter-spacex-Data%20wrangling.ipynb)

EDA with Data Visualization

Primarily visualized data with scatter plots to plot to different variables with each other. For supporting visualization bar and line charts were used.

Github:

https://github.com/m150x/ibm_data-science/blob/main/capstone/edadataviz.ipynb

EDA with SQL

- Unique launch site names
- Query for particular launch sites by their names
- Total payload
- Average payload for specific booster
- Query for first successful landing
- Query for boosters with payload between 4000 and 6000
- Total count of successful and failed missions
- Query for boosters with maximum payload
- Query for month names
- Ranking of count of landing outcomes for date range

Github:

https://github.com/m150x/ibm_data-science/blob/main/capstone/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Markers for showing of geo locations

Clusters for grouping of markers

Circles to emphasize locations / markers

Lines for showing of distance between markers

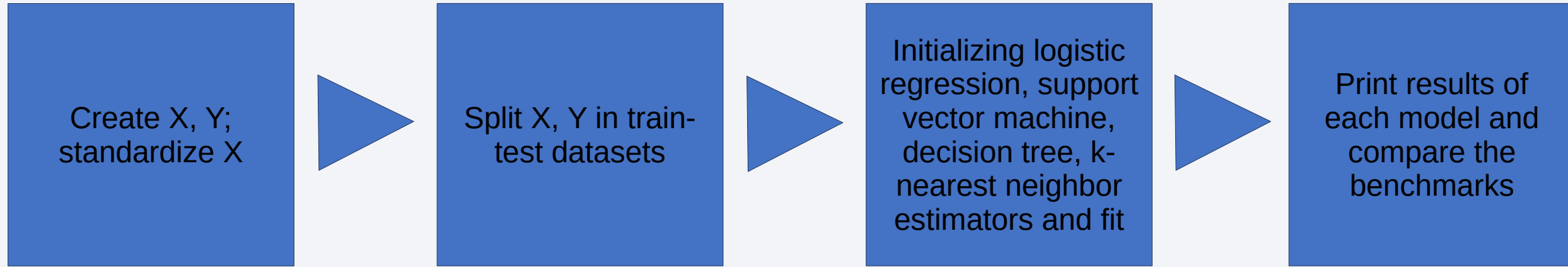
Github:

[https://github.com/m150x/ibm_data-science/blob/main/capstone/
lab_jupyter_launch_site_location.ipynb](https://github.com/m150x/ibm_data-science/blob/main/capstone/lab_jupyter_launch_site_location.ipynb)

Build a Dashboard with Plotly Dash

Summary of plots:
Bar, Line, Pie

Predictive Analysis (Classification)

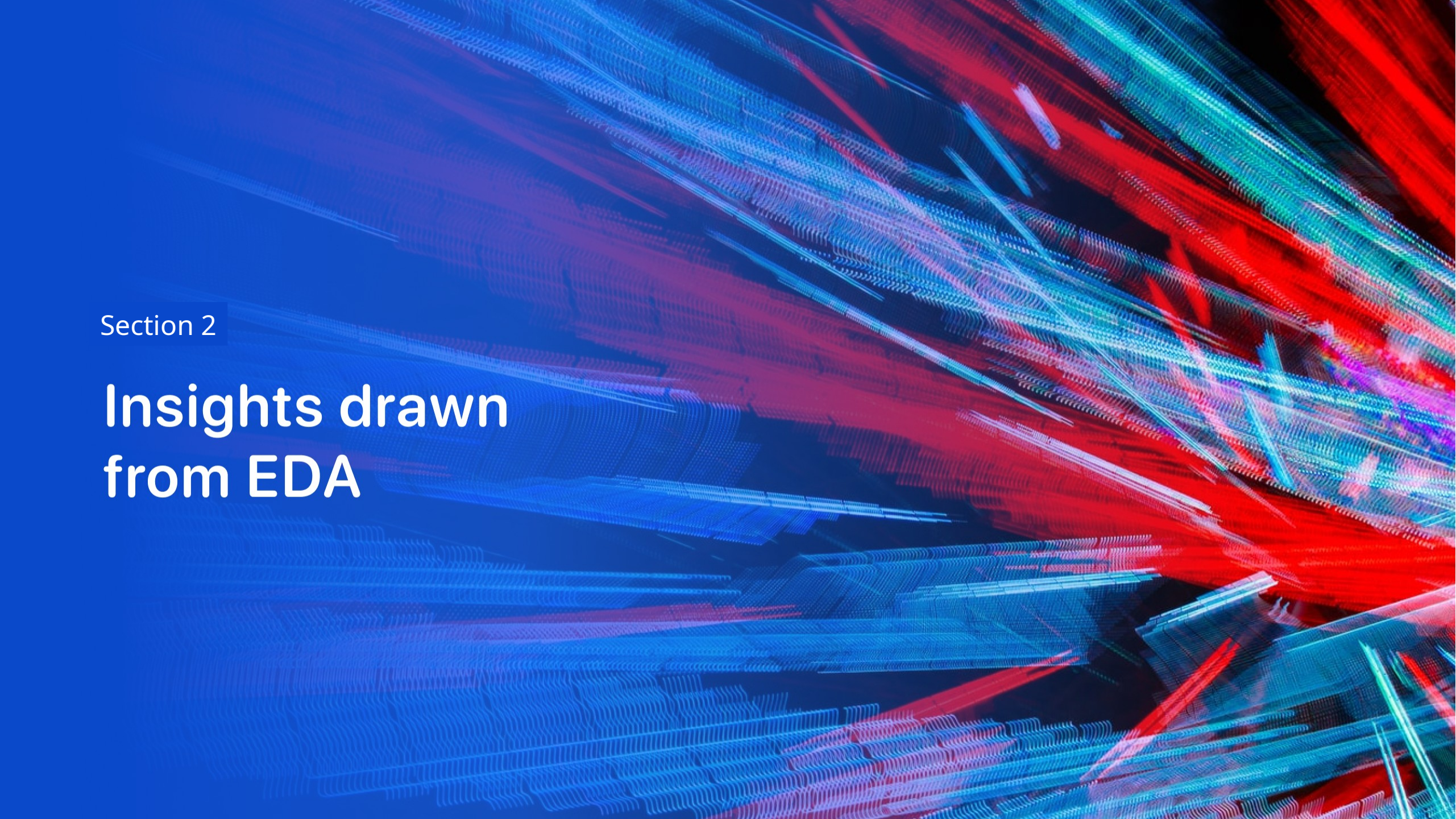


Github:

https://github.com/m150x/ibm_data-science/blob/main/capstone/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

- Exploratory data analysis results
 - Extracted data successfully via API request and web scraping
- Interactive analytics demo in screenshots
 - SQL particularly useful for data selection (queries)
 - Geographical insights due to folium and appealing interactive presentation with plotly
- Predictive analysis results
 - Similar performances with no outstanding results, decision tree performed best

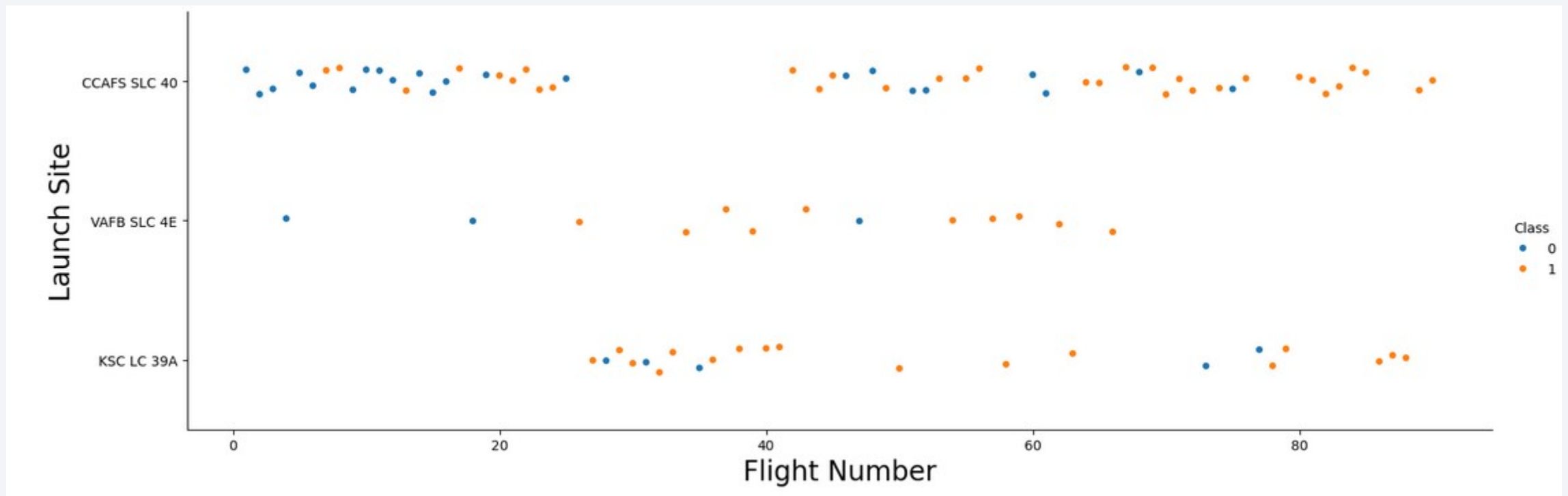
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

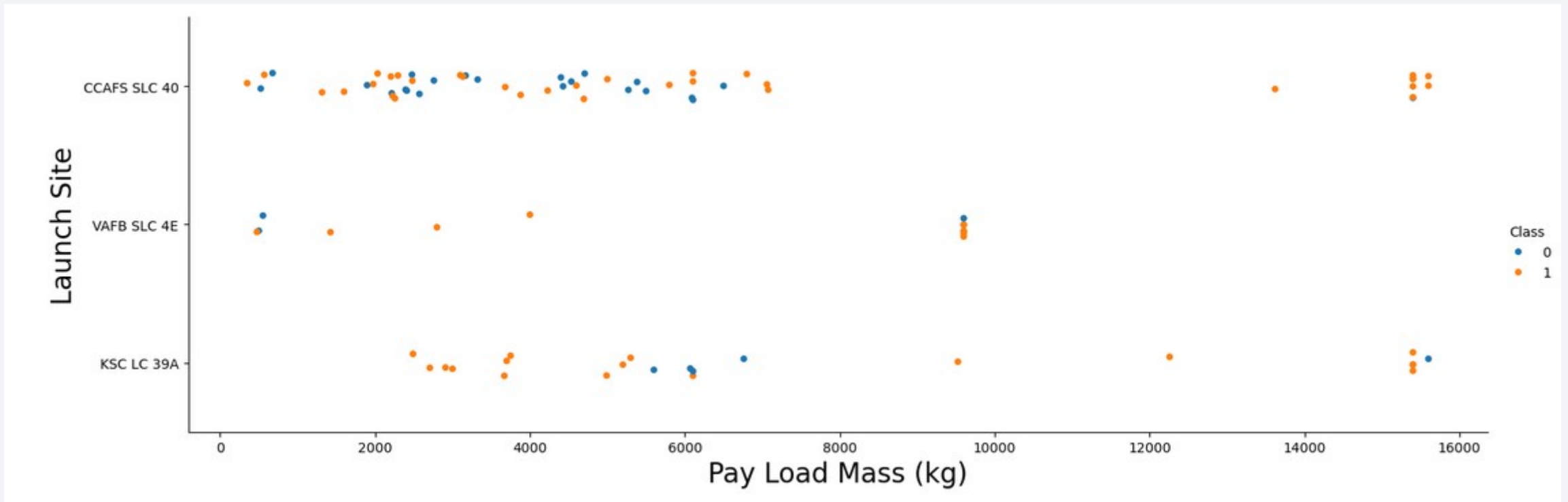
Flight Number vs. Launch Site

- No clear implications



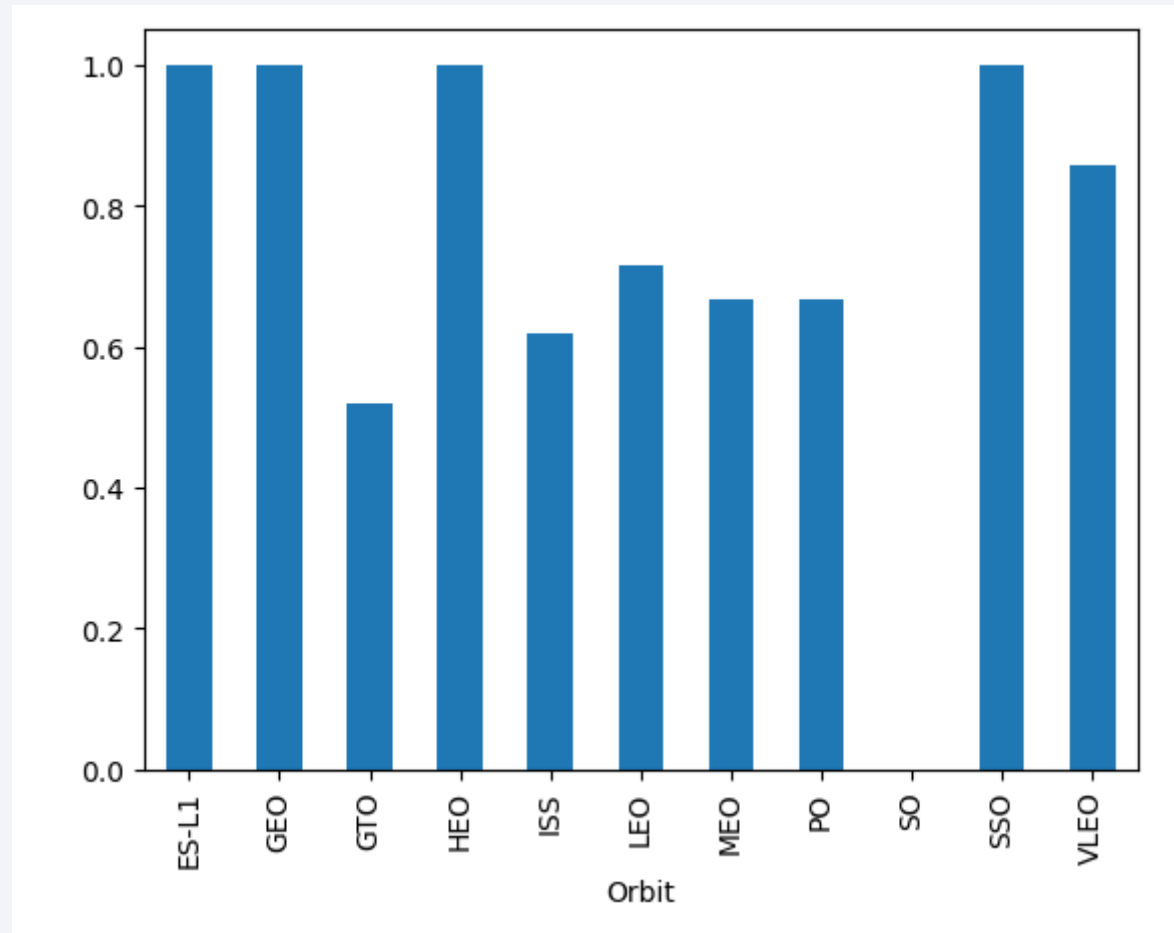
Payload vs. Launch Site

- Higher payloads tend to be more successful



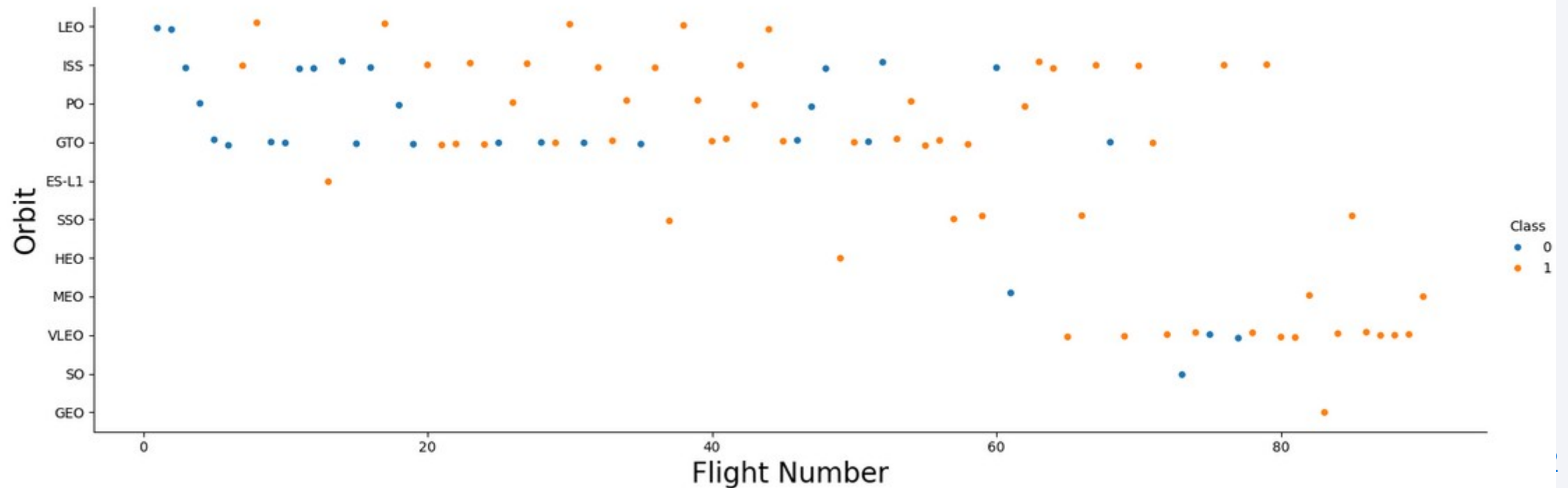
Success Rate vs. Orbit Type

- GTO performs poorly;
ES-L1, GEO, HEO, SSO
perfect score



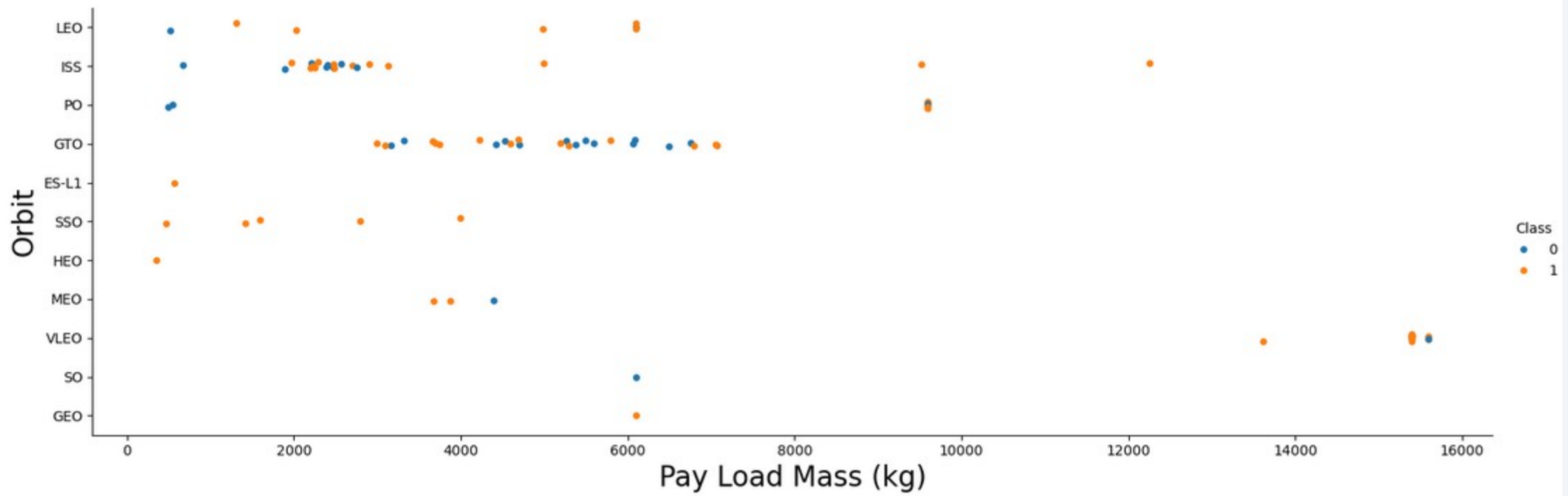
Flight Number vs. Orbit Type

- Clear improvements for LEO, slight improvements for GTO



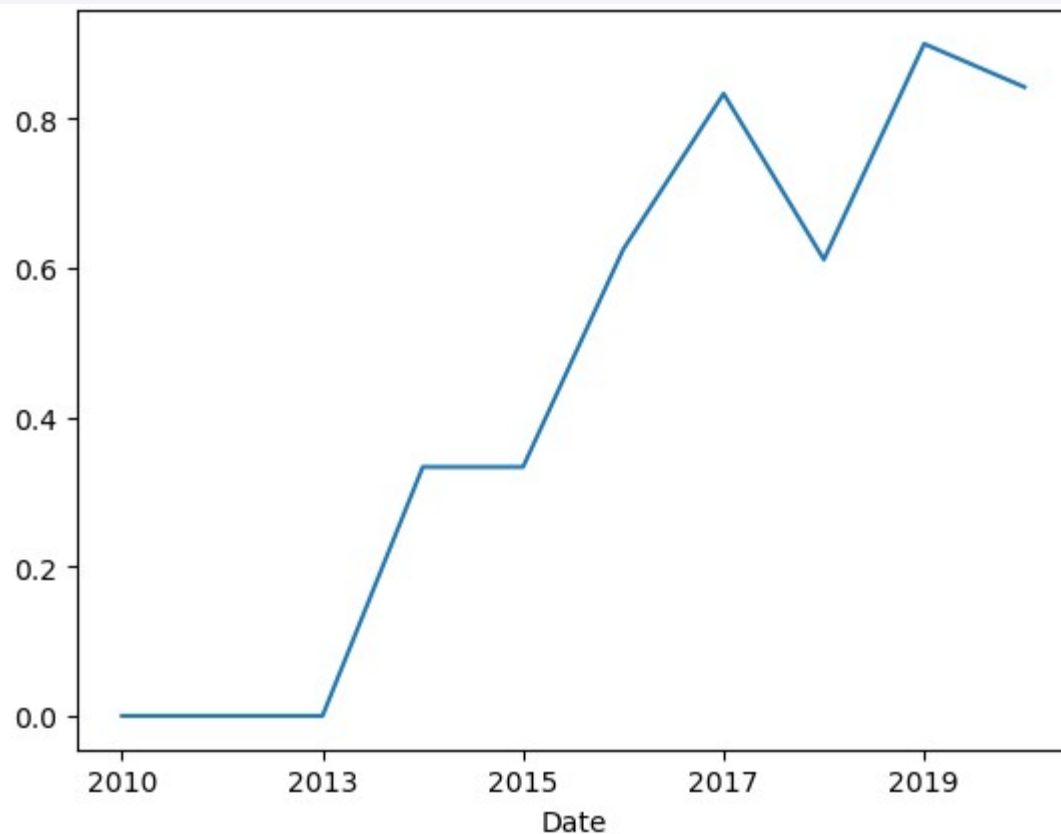
Payload vs. Orbit Type

- Higher payload seems to favor success, needs to be investigated further though



Launch Success Yearly Trend

- Clear improvements of success rates, with little bump in 2018



All Launch Site Names

Find the names of the unique launch sites

Four launch sites

: **Launch_Site**

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with `CCA`

First five records where launch site begins with CCA

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Calculate the total payload carried by boosters from NASA

Total payload amounts to 48213 kg

```
: SUM("PAYLOAD_MASS_KG_")  
-----  
48213
```

Average Payload Mass by F9 v1.1

Calculate the average payload mass carried by booster version F9 v1.1

Average payload amounts to 2928.4 kg

```
AVG("PAYLOAD_MASS_KG_")
```

2928.4

First Successful Ground Landing Date

Find the dates of the first successful landing outcome on ground pad

First successful landing on ground pad in late 2015

MIN("DATE")

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

List of names of boosters with successful landing on drone ship and Payload between 4000 and 6000

Booster_Version
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1014
F9 v1.1 B1016
F9 FT B1020
F9 FT B1022
F9 FT B1026
F9 FT B1030
F9 FT B1021.2
F9 FT B1032.1
F9 B4 B1040.1
F9 FT B1031.2
F9 B4 B1043.1
F9 FT B1032.2
F9 B4 B1040.2
F9 B5 B1046.2
F9 B5 B1047.2
F9 B5 B1046.3
F9 B5B1054
F9 B5 B1048.3
F9 B5 B1051.2
F9 B5B1060.1
F9 B5 B1058.2
F9 B5B1062.1

Total Number of Successful and Failure Mission Outcomes

Calculate the total number of successful and failure mission outcomes

Count of mission outcomes by grouping of mission outcome

Mission_Outcome	COUNT("Mission_Outcome")
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

List the names of the booster which have
carried the maximum payload mass
Maximum payload of each booster

Booster_Version	MAX("PAYLOAD_MASS_KG_")
F9 B4 B1039.2	2647
F9 B4 B1040.2	5384
F9 B4 B1041.2	9600
F9 B4 B1043.2	6460
F9 B4 B1039.1	3310
F9 B4 B1040.1	4990
F9 B4 B1041.1	9600
F9 B4 B1042.1	3500
F9 B4 B1043.1	5000
F9 B4 B1044	6092
F9 B4 B1045.1	362
F9 B4 B1045.2	2697
F9 B5 B1046.1	3600
F9 B5 B1046.2	5800
F9 B5 B1046.3	4000
F9 B5 B1046.4	12050
F9 B5 B1047.2	5300
F9 B5 B1047.3	6500
F9 B5 B1048.2	3000
F9 B5 B1048.3	4850
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.2	9600

2015 Launch Records

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Failed landing outcomes in 2015

<code>: substr("Date",6,2)</code>	<code>Landing_Outcome</code>	<code>Booster_Version</code>
01	Failure (drone ship)	F9 v1.1 B1012
04	Failure (drone ship)	F9 v1.1 B1015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Ranking of counts of landing outcomes

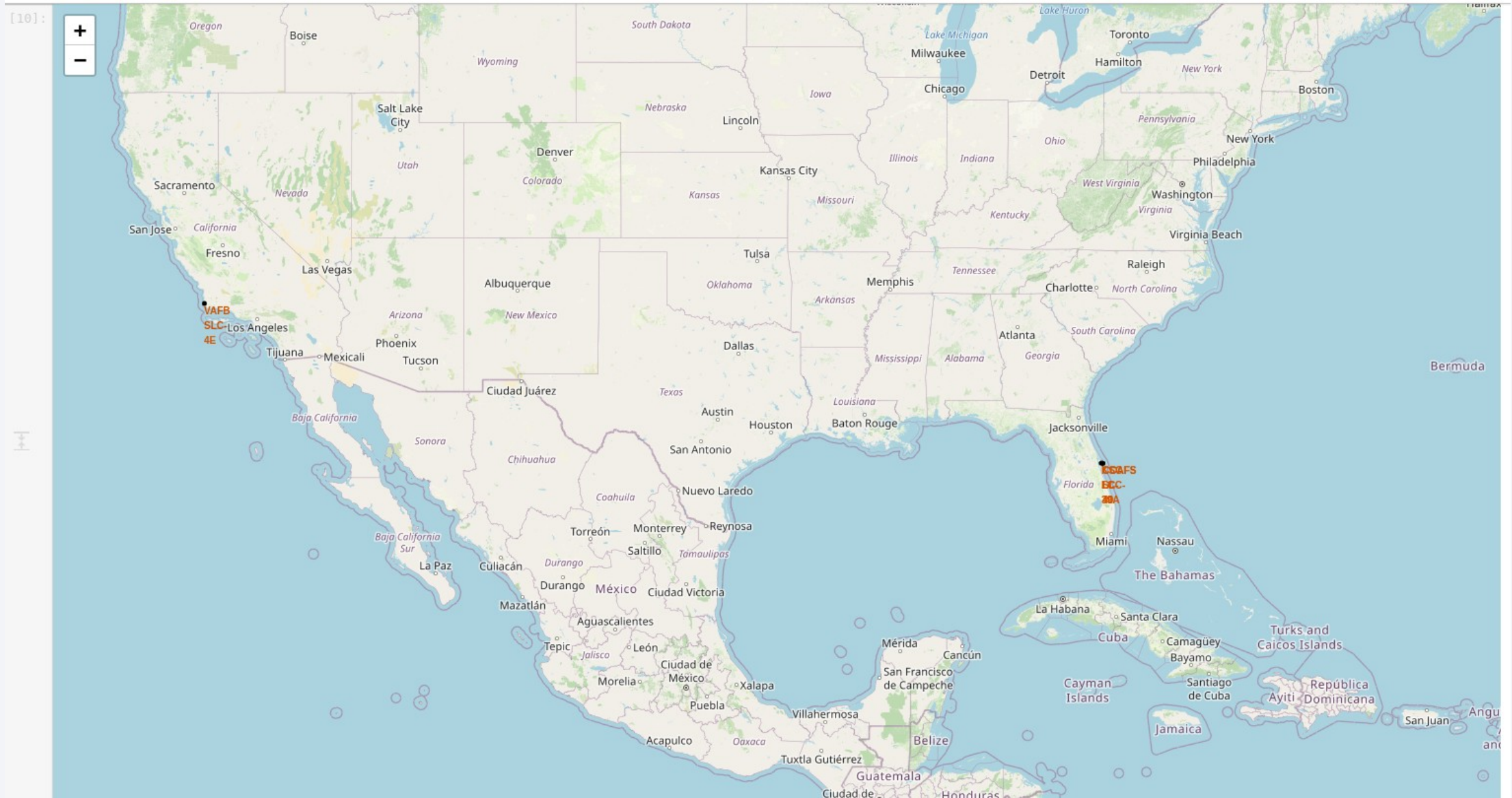
Landing_Outcome	CNT
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The background is a deep blue gradient.

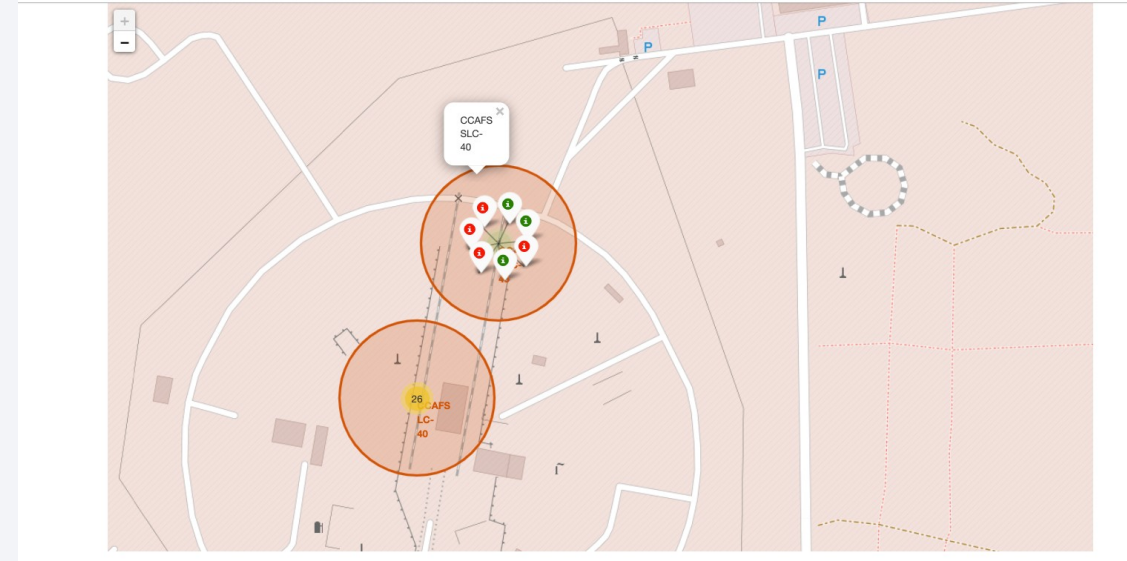
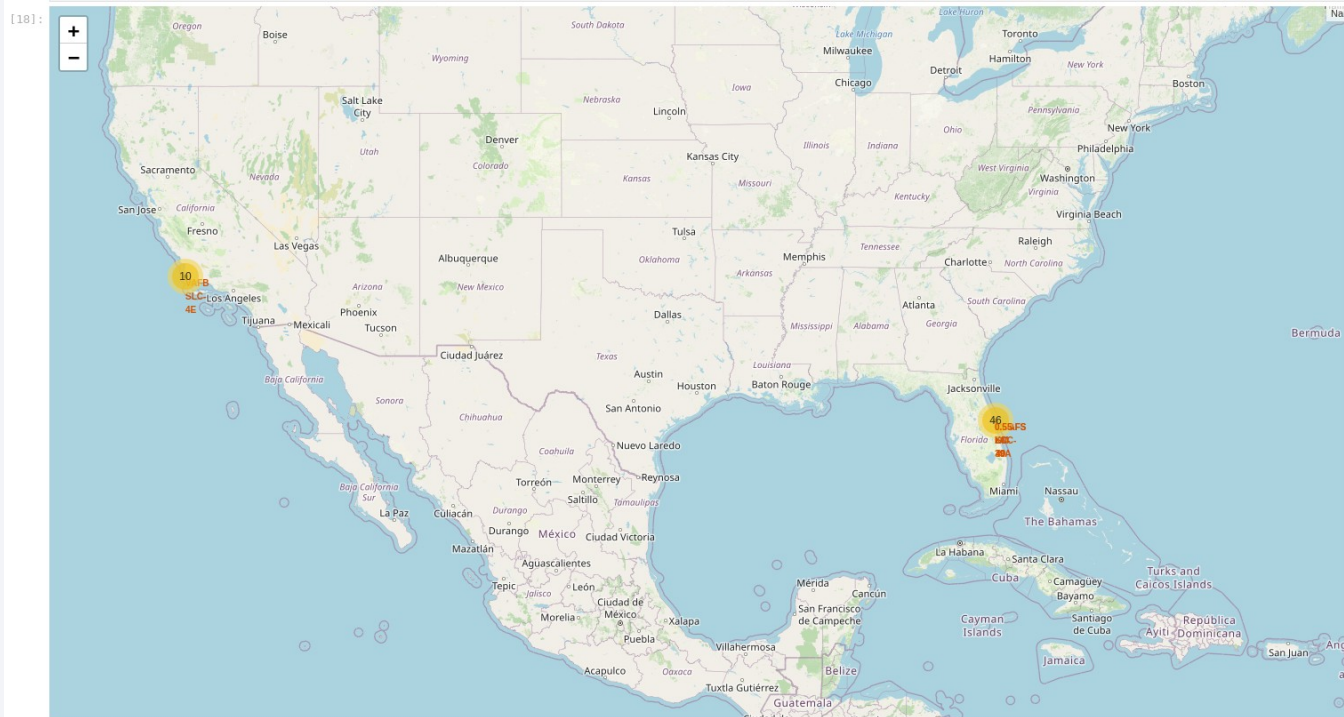
Section 3

Launch Sites Proximities Analysis

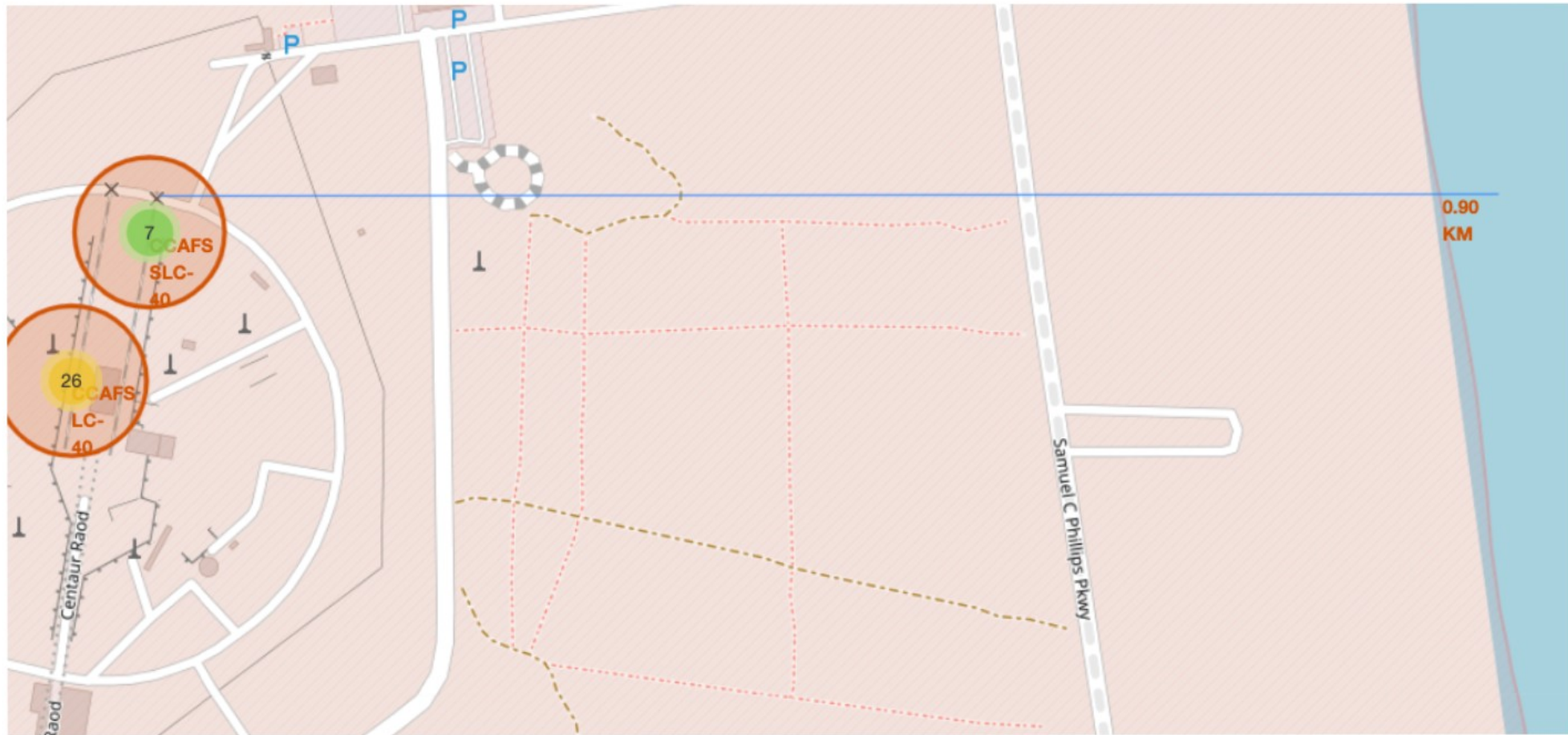
Launch Site Locations

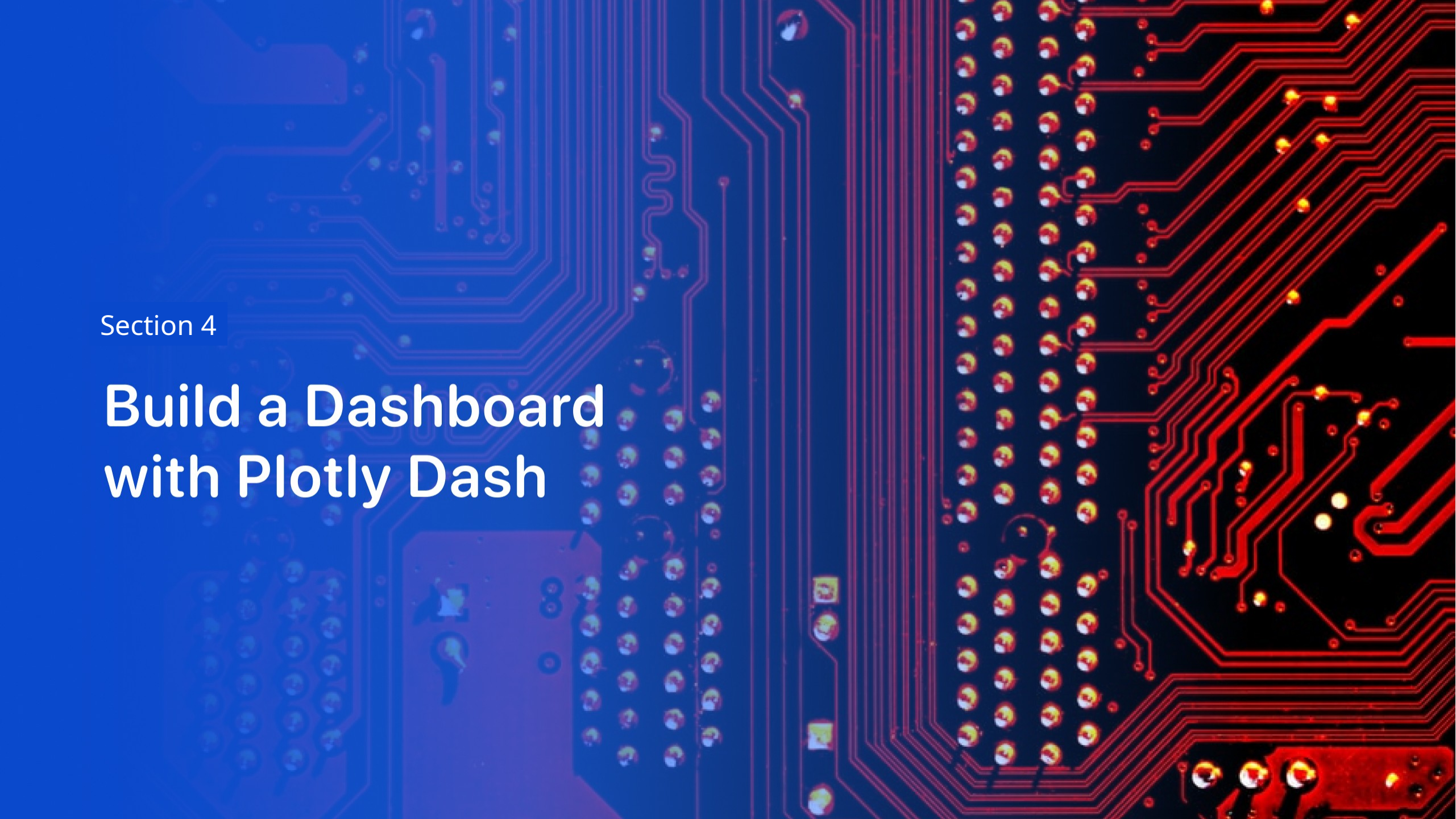


Launch Site Location Clusters and In-Depth View



Launch Site Location Proximities

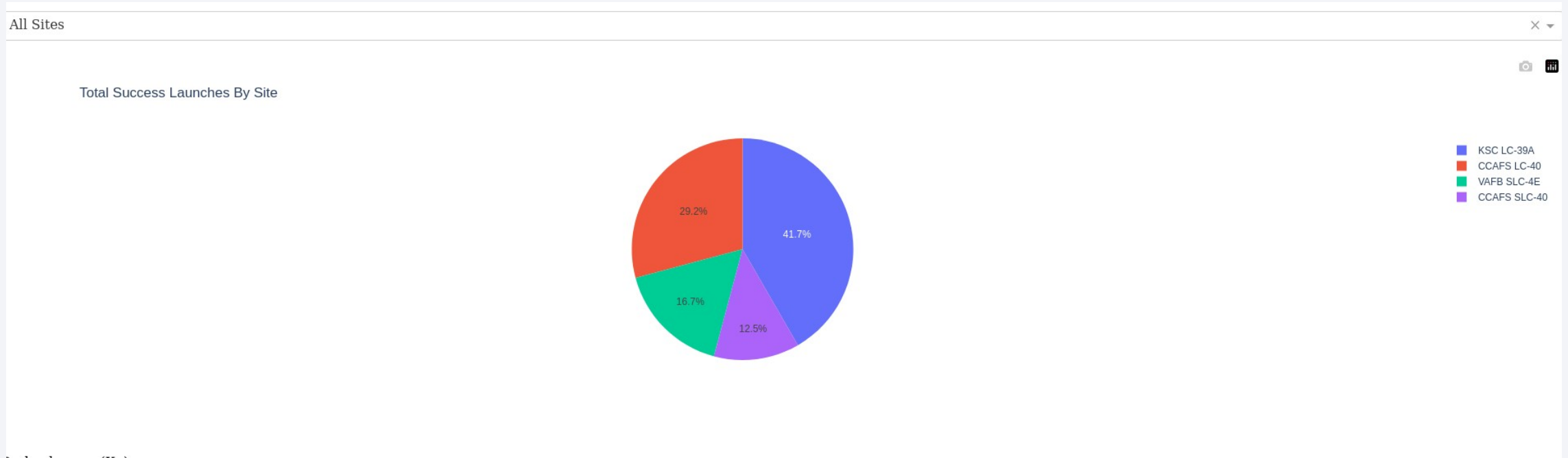




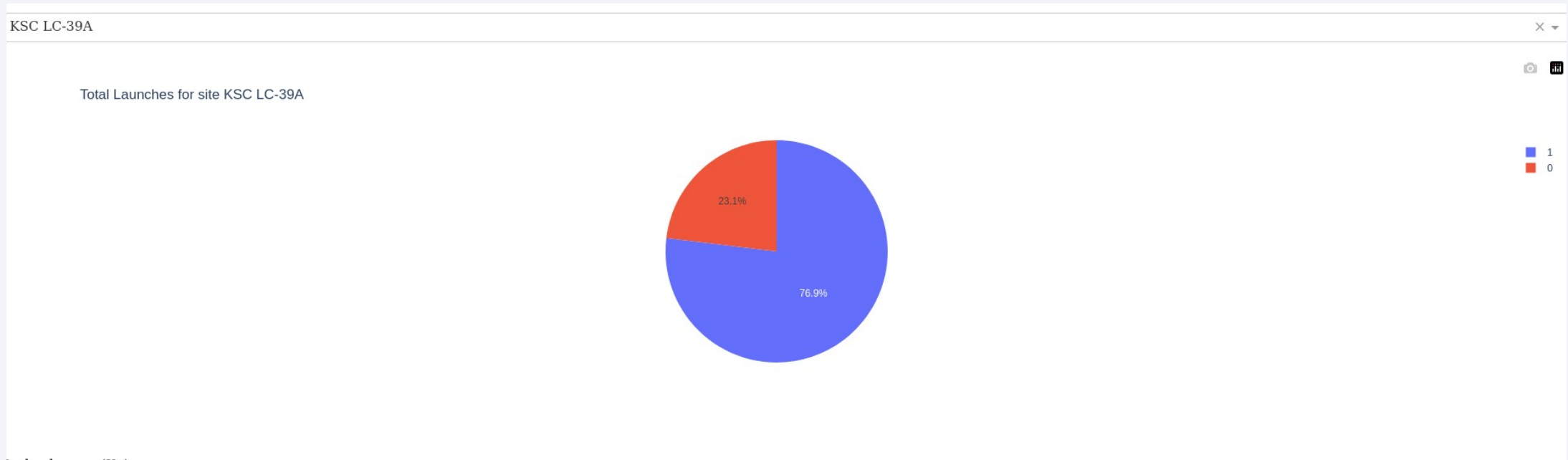
Section 4

Build a Dashboard with Plotly Dash

Success overview for all launch sites



Success/Failure share for best launch site

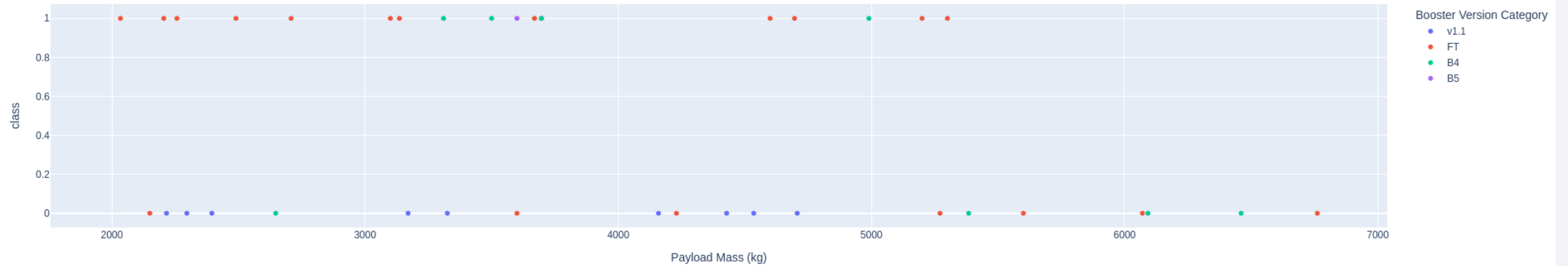


Scatter for plotting Payload vs Success

Payload range (Kg):

0 100

All sites - payload mass between 2,000kg and 7,000kg





Section 5

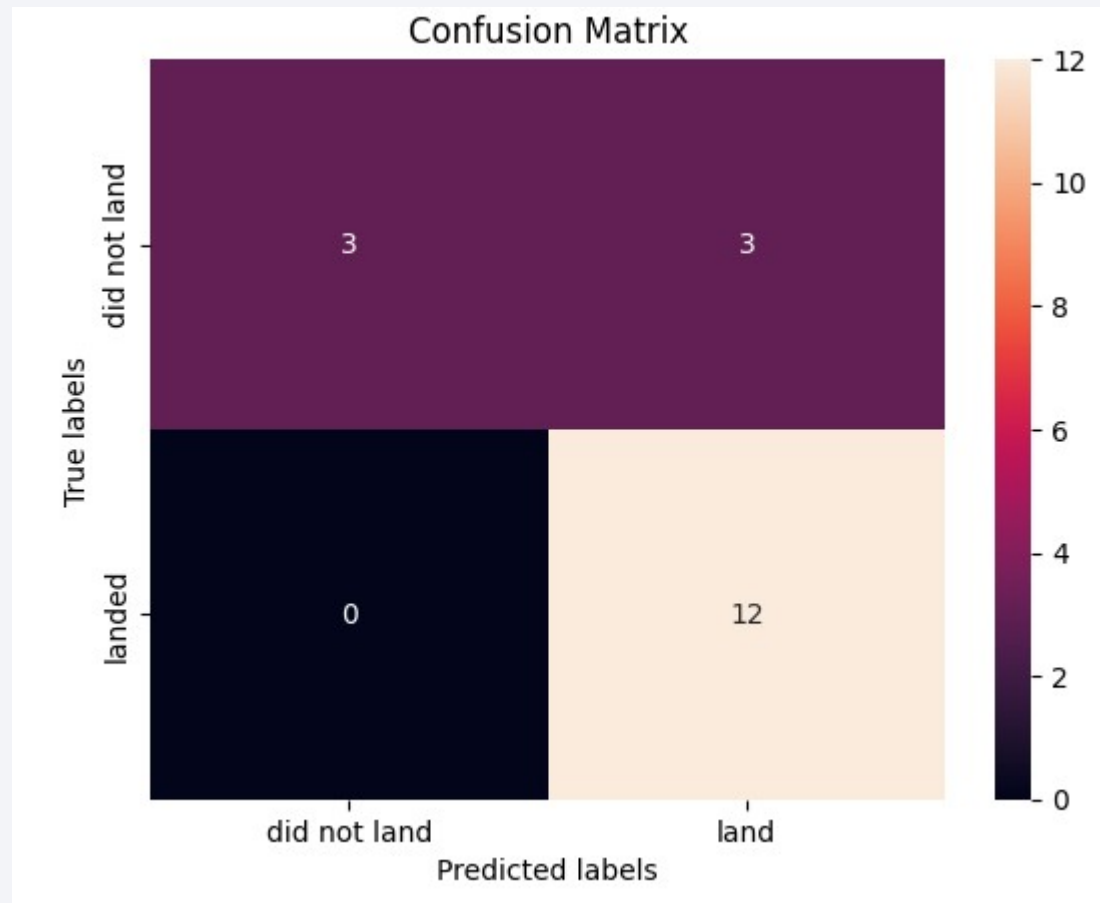
Predictive Analysis (Classification)

Classification Accuracy

- All models had an accuracy of app 83%

Confusion Matrix

- All models produced the same confusion matrix; where there are 3 false negatives and 0 false positives



Conclusions

- More flights has a positive impact on success rate
- More payload seems to have a positive impact on success rate
- ES-L1, GEO, SSO, VLEO, HEO had the highest success rates
- Launch site KSC LC-39A has the highest success rate
- Machine learning models need more training data (sample to small)

Appendix

Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

